

James Hutchins

30 Colborne St. Elora Ontario, N0B1S0
519-827-5881; james.hutchins2@outlook.com
Portfolio Site: www.jameshutchins.ca

EDUCATION

Western University, London, ON, Canada Graduated 2024
Bachelor of Engineering Science, Mechanical, and materials engineering

- Dean's Honour List

Western University, London, ON, Canada Graduated 2025
Bachelor of Science, Computer Science

- Dean's Honour List

PROJECTS

Please see my portfolio website: [James Hutchins](http://JamesHutchins.com)

WORK EXPERIENCE

Zenith Vision Analytics, Ontario, Canada — 2024-Present
Founder

- Marketing data collection and presentation services focused on the pool service industry.
- Built and operated continually running Apache Airflow pipelines on AWS for data collection, data quality validation, and refresh scheduling.
- Designed and developed multiple SQL data stores in PostgreSQL to support ingestion, enrichment, and analytics workloads.
- Created and maintained property and geospatial datasets by collecting, cleaning, and organizing data from multiple sources, enabling consistent property-level insights.
- Processed TB-scale imagery datasets using computer vision models (YOLO, SAM) to detect pools and generate structured outputs usable by downstream analytics.
- Implemented PostgreSQL vector-based search to power product/pricing comparison workflows and improve relevance in matching/lookup tasks.
- Delivered stakeholder-ready BI outputs (dashboards/semantic layers) that translated collected data into actionable recommendations and ROI clarity.

Bank Of Montreal, Toronto, Ontario, Canada — Winter 2025
Data Analyst

- Supported data quality analytics operations in Technology & Operations, partnering with internal stakeholders on reporting and issue triage.
- Built Python + Azure ETL pipelines with validation checks and data-quality rules to improve reliability of downstream datasets.
- Developed Power BI dashboards and SQL reporting to surface KPIs, anomalies, and operational trends for business partners.
- Automated repetitive workflows with cloud scripting, reducing manual effort, and accelerating recurring deliverables.

- Contributed to data quality validation pipelines processing ~500M rows/day, focusing on robustness, monitoring, and maintainability.

Dialogix Services Inc, Toronto, Ontario, Canada — June 2025-Present
Co-Founder

- Built and standardized CI/CD pipelines to ship changes faster and more reliably across environments.
- Implemented automated testing and quality gates to reduce regressions and increase release confidence.
- Implemented error detection, structured logging, storage, and reporting workflows to speed up issue discovery during ongoing support.
- Co-authored and shipped a production application used by leading educational institutions (including the Ivey Business School).
- Maintained a live production codebase: triaged issues, executed QA, reviewed pull requests, and delivered bug fixes and feature releases on a regular cadence.

Blue Guardian, London, Ontario, Canada — Winter 2023
Machine Learning NLP R&D Intern

- Developed a BERT-based NLP pipeline to detect early risk signals from speech-pattern text analysis for a parental monitoring mobile app.
- Improved classification performance by benchmarking statistical inference approaches and traditional ML baselines (regression, decision trees) against transformer-based outputs.
- Created and curated annotated datasets, and documented labeling guidelines and feature engineering to support retraining, reproducibility, and quality control.

Vector Institute, London, Ontario, Canada — Summer 2024
Generative AI (Artificial Intelligence) Research Intern

- Engineered PyTorch ML pipelines including dataset preparation, cleaning, and feature engineering for model training and evaluation.
- Leveraged transformer-based approaches and diffusion models (e.g., Stable Diffusion) to generate “seen” images from participants’ EEG signals, including building/iterating on data cleaning systems to make signals model-ready.
- Contributed to model development and experimental evaluation in a high-rigor research environment (clear metrics, reproducible runs).
- Managed large-scale training runs on compute clusters, including run configuration, monitoring, and results tracking.
- Collaborated with researchers to iterate quickly on hypotheses, baselines, and ablation-style comparisons.

Scotiabank, Toronto, Ontario, Canada — Summer 2023
Small Business Analyst

- Performed financial risk assessments by analyzing cash flow, collateral, industry trends, and client documentation.
- Improved data quality and consistency for credit pricing reporting by implementing validation processes and tighter documentation standards.

- Built VBA-based automation to standardize customer assessments, reduce manual effort, and improve repeatability across analysts.
- Produced pricing recommendations and competitive research briefs to support underwriting and strategic decisions.